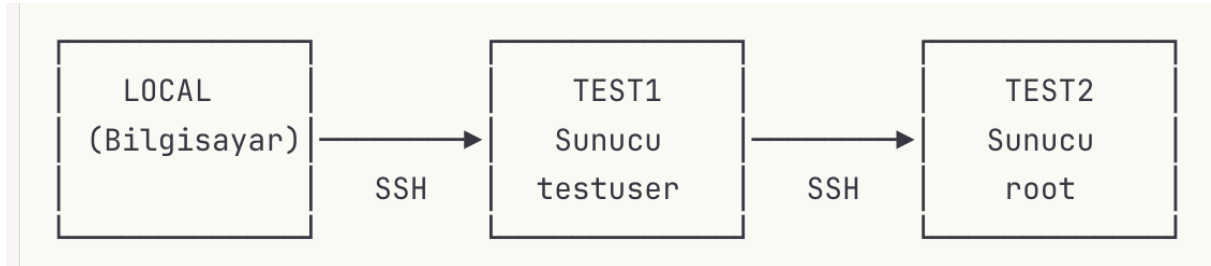


SSH KEY SETUP - PASSWORDLESS LOGIN

Senaryo:

Bu lab'da test1 ve test2 adında 2 sunucu oluşturacağız. Test1 sunucuda testuser adında yeni kullanıcı oluşturup, hem SSH key ile hem de parola ile localden test1 sunucusuna nasıl bağlanılacağını göstereceğiz. Son olarak test1 sunucudan test2 sunucusuna ssh key ile şifresiz bağlantı nasıl yapılır göstereceğiz..



Bağlantı Akışı:

1. LOCAL → TEST1 (SSH Key ile)
2. LOCAL → TEST1 (Password ile)
3. TEST1 → TEST2 (SSH Key ile - Passwordless)

test1: 100.26.43.30

test2: 54.221.93.78

1) test1'de **testuser** oluştur

Sunucu: **test1**

test1'e yönetici hesabınla bağlan:

```
ssh -i ec2_ssh_key.pem ubuntu@100.26.43.30
```

Kullanıcıyı oluştur:

```
sudo adduser testuser
```

```
ubuntu@ip-172-31-25-151:~$ sudo adduser testuser
info: Adding user `testuser' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `testuser' (1001) ...
info: Adding new user `testuser' (1001) with group `testuser (1001)' ...
info: Creating home directory `/home/testuser' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for testuser
Enter the new value, or press ENTER for the default
    Full Name []:
    Room Number []:
    Work Phone []:
    Home Phone []:
    Other []:
Is the information correct? [Y/n] y
info: Adding new user `testuser' to supplemental / extra groups `users' ...
info: Adding user `testuser' to group `users' ...
ubuntu@ip-172-31-25-151:~$
```

Doğrula:

`id testuser`

```
ubuntu@ip-172-31-25-151:~$ id testuser
uid=1001(testuser) gid=1001(testuser) groups=1001(testuser),100(users)
ubuntu@ip-172-31-25-151:~$ ls -l /home/
total 8
drwxr-x--- 2 testuser testuser 4096 Oct 12 21:05 testuser
drwxr-x--- 4 ubuntu   ubuntu   4096 Oct 12 21:05 ubuntu
ubuntu@ip-172-31-25-151:~$
```

2) Lokal makinede SSH anahtarı üret

SSH Key Pair Oluştur

```
ssh-keygen -t ed25519 -C "workshop@example.com" -f workshop_key
```

```
(base) ssh-keygen ➡ ssh-keygen -t ed25519 -C "workshop@example.com" -f workshop_key
Generating public/private ed25519 key pair.
Enter passphrase for "workshop_key" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in workshop_key
Your public key has been saved in workshop_key.pub
The key fingerprint is:
SHA256:scr2YrSj1Wwh3xu0IdTZ7ykmk2/ncRgsIGQWGMzazqM workshop@example.com
The key's randomart image is:
+--[ED25519 256]--+
|  o.o=.          |
|    ++          |
|  o  . o        |
| . .  o * .     |
|  o  o S o o    |
|   +o.* o o o   |
| . o*.*.o + .   |
| E  o=+++++.+   |
|   .o oo*+++    |
+-----[SHA256]-----+
```

Oluşan dosyaları kontrol et

```
ls -l
```

```
(base) ssh-key 👉 ls -l
total 16
-rw----- 1 hakan  staff  411 Oct 12 22:37 workshop_key
-rw-r--r-- 1 hakan  staff  102 Oct 12 22:37 workshop_key.pub
```

Public key içeriğini gör

```
cat workshop_key.pub
```

```
(base) Downloads 👉 cat workshop_key.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIF30ncMssDdojGJEbH5LE7UT176XdeAYxIAMwFNZbyKr workshop@example.com
```

3) Public key'i test1'e ekle

Sunucuya scp komutu ile public key dosyasını testuser kullanıcısı home dizinine kopyala

```
scp -i ec2_ssh_key.pem workshop_key.pub
ubuntu@100.26.43.30:/home/testuser/
```

Sunucuya ubuntu user ile bağlan

```
ssh -i ec2_ssh_key.pem ubuntu@100.26.43.30
```

```
(base) Downloads 👉 scp -i ec2_ssh_key.pem workshop_key.pub ubuntu@100.26.43.30:/home/ubuntu
workshop_key.pub 100% 102 0.6KB/s 00:00
(base) Downloads 👉 ssh -i ec2_ssh_key.pem ubuntu@100.26.43.30
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1011-aws x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Sun Oct 12 21:35:25 UTC 2025
```

testuser kullanıcısı için .ssh klasörünü oluştur ve gerekli yetkileri ver

```
sudo -u testuser mkdir -p ~testuser/.ssh && sudo chmod 700
~testuser/.ssh
sudo chown -R testuser:testuser ~testuser/.ssh
```

```
ubuntu@ip-172-31-25-151:~$ sudo ls -la /home/testuser/
total 20
drwxr-x--- 2 testuser testuser 4096 Oct 12 21:29 .
drwxr-xr-x 4 root     root     4096 Oct 12 21:05 ..
-rw-r--r-- 1 testuser testuser  220 Oct 12 21:05 .bash_logout
-rw-r--r-- 1 testuser testuser 3771 Oct 12 21:05 .bashrc
-rw-r--r-- 1 testuser testuser  807 Oct 12 21:05 .profile
ubuntu@ip-172-31-25-151:~$ sudo -u testuser mkdir -p ~testuser/.ssh && sudo chmod 700 ~testuser/.ssh
ubuntu@ip-172-31-25-151:~$ sudo ls -la /home/testuser/
total 24
drwxr-x--- 3 testuser testuser 4096 Oct 12 21:30 .
drwxr-xr-x 4 root     root     4096 Oct 12 21:05 ..
-rw-r--r-- 1 testuser testuser  220 Oct 12 21:05 .bash_logout
-rw-r--r-- 1 testuser testuser 3771 Oct 12 21:05 .bashrc
-rw-r--r-- 1 testuser testuser  807 Oct 12 21:05 .profile
drwx----- 2 testuser testuser 4096 Oct 12 21:30 .ssh
```

testuser için authorized_keys dosyasını oluştur

sudo komutunu kullanarak testuser için **authorized_keys** dosyasını oluşturduktan sonra daha önce ubuntu kullanıcı dizinine kopyaladığımız dosyayı ~testuser/.ssh/authorized_keys içine ekleyeceğiz

```
sudo su
cd /home/testuser/
ls -la
cd .ssh/
cat /home/ubuntu/workshop_key.pub >> authorized_keys
cat authorized_keys
```

Sonra .ssh testuser için authorized_keys dosyasına erişim izinleriniz ayarlayacağız.

```
sudo chmod 600 ~testuser/.ssh/authorized_keys && sudo chown
testuser:testuser ~testuser/.ssh/authorized_keys
```

```

ubuntu@ip-172-31-25-151:~$ sudo su
root@ip-172-31-25-151:/home/ubuntu# cd /home/testuser/
root@ip-172-31-25-151:/home/testuser# ls -l
total 0
root@ip-172-31-25-151:/home/testuser# ls -la
total 24
drwxr-x--- 3 testuser testuser 4096 Oct 12 21:30 .
drwxr-xr-x 4 root      root    4096 Oct 12 21:05 ..
-rw-r--r-- 1 testuser testuser 220 Oct 12 21:05 .bash_logout
-rw-r--r-- 1 testuser testuser 3771 Oct 12 21:05 .bashrc
-rw-r--r-- 1 testuser testuser 807 Oct 12 21:05 .profile
drwx----- 2 testuser testuser 4096 Oct 12 21:50 .ssh
root@ip-172-31-25-151:/home/testuser# cd .ssh/
root@ip-172-31-25-151:/home/testuser/.ssh# ls -l
total 0
root@ip-172-31-25-151:/home/testuser/.ssh# cat /home/ubuntu/workshop_key.pub >> authorized_keys
root@ip-172-31-25-151:/home/testuser/.ssh# cat authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIF30ncMssDdojGJEbH5LE7UT176XdeAYxIAMwFNZbyKr workshop@example.com
root@ip-172-31-25-151:/home/testuser/.ssh# sudo chmod 600 ~testuser/.ssh/authorized_keys && sudo chown testuser:testuser ~testuser/.ssh/authorized_keys
root@ip-172-31-25-151:/home/testuser/.ssh# ls -la
total 12
drwx----- 2 testuser testuser 4096 Oct 12 21:52 .
drwxr-x--- 3 testuser testuser 4096 Oct 12 21:30 ..
-rw----- 1 testuser testuser 102 Oct 12 21:52 authorized_keys
root@ip-172-31-25-151:/home/testuser/.ssh#

```

4) Lokal → test1 bağlan (anahtarla ve parolayla)

4.a) Anahtarla (parolasız)

İstemci: **lokal**

`ssh -i workshop_key testuser@TEST1_IP`

```
ssh -i workshop_key testuser@100.26.43.30
```

Başarılıysa parolasız giriş tamam.

4.b) Parolayla (karşılaştırma amaçlı)

test1 sunucusunda passwordle erişimi aktif etme

İstemci: **lokal**

Örnek: `ssh -i ec2_ssh_key testuser@TEST1_IP`

```
ssh -i ec2_ssh_key.pem ubuntu@100.26.43.30
```

ssh config dosyalarında passwordle girişi aktif etmek için vi editörünü kullanalım

```

sudo vi
/etc/ssh/sshd_config.d/60-cloudimg-settings.conf

```

ubuntu@ip-172-31-25-151: ~

PasswordAuthentication no

no yazan kısmı yes yapıp kaydedin çıkın.

ubuntu@ip-172-31-25-151: ~

PasswordAuthentication yes

Değişikliğin aktif olması için ssh servisini yeniden başlatın yada ssh config dosyasını reload edin.

```
sudo systemctl reload ssh # RHEL: sshd
```

şimdi Sunucudan çıkıp lokal bilgisayarınızdan test1 sunucusuna şifresiz girmeyi deneyin

```
(base) ~ ➔ ssh testuser@100.26.43.30
testuser@100.26.43.30's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1011-aws x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Sun Oct 12 22:11:40 UTC 2025

System load:  0.0                Processes:            123
Usage of /:   26.3% of 6.71GB    Users logged in:     1
Memory usage: 23%                IPv4 address for enX0: 172.31.25.151
Swap usage:   0%
```

5) test2 tarafını hazırla (kullanıcı + key kabulü)

5.a) test2'de testuser oluştur

Sunucu: test2ye bağlan

ssh ubuntu@TEST2_IP

```
ssh -i ec2_ssh_key.pem ubuntu@54.221.93.78
```

```
(base) Downloads ➡ ssh -i ec2_ssh_key.pem ubuntu@54.221.93.78
The authenticity of host '54.221.93.78 (54.221.93.78)' can't be established.
ED25519 key fingerprint is SHA256:wAxMr0kUye0g0LZlwNLkisbuVA1c9mbV0tvv5QAiB8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '54.221.93.78' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1011-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Oct 12 22:15:08 UTC 2025

System load:  0.0          Processes:      110
Memory usage: 6.4%        Swap: 0k/0k
Disk usage:   25.8%        Filesystems: 1
CPU state:    0%           6.6-71.68
Network:      enp0s3        0
```

user oluştur

```
sudo adduser testuser
```

```
ubuntu@ip-172-31-19-245:~$ sudo adduser testuser
info: Adding user `testuser' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `testuser' (1001) ...
info: Adding new user `testuser' (1001) with group `testuser (1001)' ...
info: Creating home directory `/home/testuser' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for testuser
Enter the new value, or press ENTER for the default
  Full Name []:
  Room Number []:
  Work Phone []:
  Home Phone []:
  Other []:
Is the information correct? [Y/n] y
info: Adding new user `testuser' to supplemental / extra groups `users' ...
info: Adding user `testuser' to group `users' ...
ubuntu@ip-172-31-19-245:~$
```

5.b) test1'de **testuser** için bir anahtar üret (test1→test2 şifresiz bağlantı için)

Sunucu: **test1** (testuser ile)

test1'e testuser olarak gir (4.a adımıyla):

```
ssh testuser@TEST1_IP
```

```
ssh -i workshop_key testuser@100.26.43.30
```

test1 üzerinde testuser'a özel anahtar üret:

```
ssh-keygen -t ed25519 -C "testuser@test1"
```

Varsayılan ~/.ssh/id_ed25519 dosyasını kullanabilir ya da isim ver.

```
testuser@ip-172-31-25-151:~$ ssh-keygen -t ed25519 -C "testuser@test1"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/testuser/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/testuser/.ssh/id_ed25519
Your public key has been saved in /home/testuser/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:rT8ILiaPINLfIGDVnQIL+A6bAYQhm4c5VoMp5EoAUPY testuser@test1
The key's randomart image is:
+--[ED25519 256]--+
|%*=...          |
|BX..= . .       |
|O+o.Eo o        |
|+=.. . .        |
|o.* S .         |
|. = . . .        |
|+ o .. ...      |
|o..+oo. ...     |
| .+o.. ..       |
+-----[SHA256]-----+
testuser@ip-172-31-25-151:~$
```

```
ls -la .ssh
```

```
testuser@ip-172-31-25-151:~$ ls -la .ssh/
total 20
drwx----- 2 testuser testuser 4096 Oct 12 22:21 .
drwxr-x--- 4 testuser testuser 4096 Oct 12 22:02 ..
-rw----- 1 testuser testuser  102 Oct 12 21:52 authorized_keys
-rw----- 1 testuser testuser  411 Oct 12 22:21 id_ed25519
-rw-r--r-- 1 testuser testuser   96 Oct 12 22:21 id_ed25519.pub
testuser@ip-172-31-25-151:~$
```

Public key içeriğini copy paste ile test2 sunucusundaki .ssh/authorized_keys dosyası içine ekleyeceğiz.

```
cat /home/testuser/.ssh/id_ed25519.pub
```

```
testuser@ip-172-31-25-151:~$ cat /home/testuser/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAILQN1391LxAlFKUhme5QZPrR9BQZ96Sq0a/0iDTmt91o testuser@test1
```

5.c) test1'deki public key'i test2'deki testuser'a ekle

test1 sunucusundan `cat /home/testuser/.ssh/id_ed25519.pub` komutu ile aldığımız içeriği test2 sunucusundaki `/home/testuser/.ssh/authorized_keys` dosyası içine kopyalayacağız.

```
sudo su
cd /home/testuser/
mkdir -p .ssh && chmod 700 .ssh
sudo chown -R testuser:testuser ~testuser/.ssh
vi .ssh/authorized_keys
```

```
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAILQN1391LxA1FKUhme5QZPrR9BQZ96Sq0a/OiDTmt91o testuser@test1
```

6) test1 → test2 şifresiz bağlantıyı test et

Sunucu: **test1** (testuser ile)

`ssh testuser@TEST2_IP`

```
ssh testuser@54.221.93.78
```

Parola sormadan giriş olmalı.

```
testuser@ip-172-31-25-151:~$ ssh testuser@54.221.93.78
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1011-aws x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Sun Oct 12 22:48:49 UTC 2025

System load:  0.0          Processes:           126
Usage of /:   26.3% of 6.71GB Users logged in:       1
Memory usage: 23%         IPv4 address for enX0: 172.31.19.245
Swap usage:   0%
```

7) Kısa adlar için SSH config

İstemci:test1

```
vi ~/.ssh/config
Host test2
  HostName 54.221.93.78
  User testuser
  IdentityFile ~/.ssh/id_ed25519
```

Kullanım:

ssh test2

```
testuser@ip-172-31-25-151:~$ ssh test2
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1011-aws x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Sun Oct 12 22:57:21 UTC 2025

System load:  0.06               Processes:            127
Usage of /:   26.3% of 6.71GB    Users logged in:     1
Memory usage: 23%               IPv4 address for enX0: 172.31.19.245
```